PROTOCOL DESIGN / PORTALDOT V3

One asset.
Two runtimes.
One enforceable title.

LEMO gives one asset one revisioned title: a legal record on the native side and a market representation on EVM. The design requires each face to verify the other before it moves.

REVIVE IS LIVE TODAY. THE WASM CONTRACTS MODULE IS LISTED FOR V3.0, NOT DEPLOYED.

LEGAL FACE	MARKET FACE	ENFORCEMENT
WASM module listed	Revive, live	Bidirectional proof

Live chain	PortaldotV3 Test
Best head	292,823
Finalized head	292,820
Finality lag	3 BLOCKS
Revive chain id	420420777

Read from the public JSON-RPC endpoints
2026-11-02 14:20 UTC / heads advance with every slot

THE GAP

Revive is live. The WASM contract module is only listed.

Portaldot V3 runs EVM contracts through Revive today. The WASM contracts module is listed for V3.0 but is not live or documented. Native assets, identity, vesting and proxy or multisig records already exist in the runtime.



1 NETWORK / 2 EXECUTION MODELS / 0 AUTOMATIC CROSS-READS

GENESIS 0xd4f9acbc…b155283bd

> When the module lands, the two will share validators and finality but not contract storage. **A proof has to cross, because a call cannot.**

THE CONSEQUENCE

Without a proof that crosses, the two halves cannot see each other.

Four failures follow from one missing piece. If no runtime can prove the other's state, no ownership rule can cross the boundary.

01

Ownership stops at the boundary

Native ownership rules do not automatically follow an ERC-20 or ERC-721 representation.

02

EVM apps cannot read the conditions

Solidity sees a token. It cannot see the identity, vesting or permission rules that make it valid.

03

Unlocking can strand a representation

Backing can leave while tokens still circulate, unless the EVM state is checked first.

04

A sync service becomes the authority

The alternative to proofs is a privileged operator that every user has to trust.

> Wrapping moves the asset. It does not **enforce** the relationship between the backing and its representation.

THE IDEA

Give the asset one title. Then give it two faces that must agree.

WASM FACE / LEGAL RECORD / MODULE LISTED	SPECIMEN
Native backing	100.00 POT
Owner	5Grw…utQY
Identity	verified
Vesting	24 months
Permissions	pure account
Revision	07

WASM → EVM PROOF

TITLE IDENTIFIER

0x7f3a…c19e

revision 07

SPECIMEN

EVM → WASM PROOF

EVM FACE / MARKET / REVIVE TODAY	SPECIMEN
Standard	ERC-20
Contract	0xA19c…B3C5
Supply	100.00 LEMO
Holder	0x8C4e…2f19
Accepted revision	07
Transfer rule	proof required

The hash is a commitment, not a proof. It names the relationship. It cannot show that backing exists or that identity holds.

The design requires each runtime to verify the other. A relayer can carry proof bytes, not authorize them. Today the root source is a trusted finalized checkpoint.

SPECIMEN VALUES. NO TITLE IS DEPLOYED ON THE TESTNET.

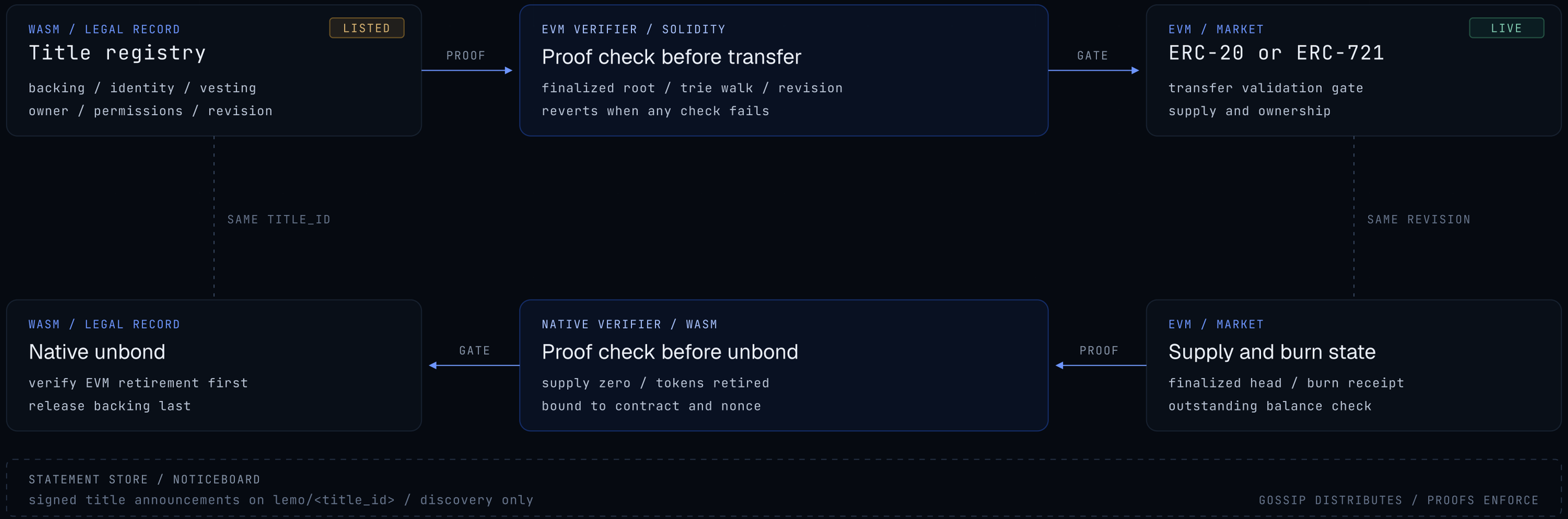
ARCHITECTURE

The design: two proofs, two gates, one title.

No shared storage and no relayer in the trust path. The EVM gate is buildable today. The native gate needs the WASM contracts module.

ARROWS CARRY PROOF BYTES, NOT CALLS

SOLID: ENFORCEMENT PATH / DASHED: DISCOVERY



STATEMENT STORE / NOTICEBOARD

signed title announcements on lemo/<title_id> / discovery only

GOSSIP DISTRIBUTES / PROOFS ENFORCE

RULE ONE

An EVM transfer needs a finalized, current native proof.

Before any ERC-20 or ERC-721 transfer settles, the EVM face proves that the native title still authorizes it.

THE PROOF MUST ESTABLISH

01 The title exists at the proven state root.

02 Backing is still bound to that title.

03 Native identity conditions are still satisfied.

04 Vesting still permits the movement.

05 The revision is current, and the proof head is finalized and fresh.

This is the design rule, not a deployed check. A valid signature over stale state is not a valid proof.

ALLOW — transfer settles

All five checks pass against one finalized native root, and the proven revision equals the revision the token last accepted.

REVERT — supply untouched

Identity cleared, backing unbound, vesting expired, proof stale, or proof from revision r-1. The transfer does not settle.

VALID PROOF | OTHERWISE REVERTED | NATIVE INPUT NEEDS THE MODULE

RULE TWO

Native backing cannot unlock while a representation is outstanding.

Before unbonding, the WASM face proves that the EVM side is fully retired at a finalized head. Burn first, unlock second.

STEP 01
Representation burned
supply to zero

→

STEP 02
Burn finalizes
finalized EVM head

→

STEP 03
EVM proof checked
in the WASM verifier

→

STEP 04
Backing unlocks
last, never first

Design rule, not a deployed check. Step 03 runs on chain only once the WASM contracts module is live; today it is listed for V3.0.

THE PROOF MUST ESTABLISH

01 ERC-20 supply is zero, or every ERC-721 representation is retired.

02 The proof belongs to this title, this contract and this operation.

03 The EVM state is finalized, the proof is current and cannot be replayed.

UNLOCK — backing released

Nothing is outstanding and the retirement proof is bound to this title at the revision being released.

REVERT — unbond refused

Any ERC-20 balance or live ERC-721 representation remains, or the proof is from another contract, revision or chain.

LIFECYCLE

Five steps. Every one leaves a receipt the other side can verify.

01

Create

Native title written and backing locked. Ownership, identity and vesting recorded on WASM.

REVISION	+1
PROOF	none yet
ENFORCED	WASM registry

02

Represent

ERC-20 or ERC-721 issued against the same title, only after the native backing is proven.

REVISION	+1
PROOF	native title
ENFORCED	EVM verifier

03

Transfer

Ownership moves through familiar token interfaces. Each transfer re-checks the title.

REVISION	unchanged
PROOF	native, current
ENFORCED	EVM gate

04

Update

Identity, backing or owner changes. The revision increments and older proofs expire.

REVISION	+1
PROOF	new revision
ENFORCED	WASM registry

05

Redeem

Representation returned and burned. Once the burn finalizes, the backing unlocks.

REVISION	+1 at burn
PROOF	EVM retirement
ENFORCED	native verifier

PROOF LINEAGE rev 06 ~~rev 06 proof~~ — rev 07 A CHANGE TO IDENTITY, BACKING OR OWNERSHIP INCREMENTS THE REVISION AND KILLS EVERY OLDER PROOF. VALID BYTES FROM REVISION 06 ARE STILL THE WRONG ANSWER AT 07.

DESIGN. NO STEP HAS RUN END TO END ON PORTALDOT V3.

THE HARD PART

The work is proof plumbing, not the token contract.

Five problems decide whether this protocol is real. Each one is a requirement, not a later optimization.

	PROBLEM	WHAT MUST HOLD	THE FAILURE IT PREVENTS
01	**Finality**	Verify against a finalized state root, never against a head an RPC simply reports.	A reorg erases the state that authorized a transfer after it settled.
02	**Freshness**	A proof accepted once cannot authorize a second, later action.	A title that changed is moved again on the strength of old bytes.
03	**Revision binding**	Every proof names the revision it proves. Any property change increments it.	Cleared identity or unbound backing leaves old proofs looking current.
04	**Replay protection**	Bind proof to chain id, genesis, title, runtime, contract, operation and nonce.	One proof is reused across titles, contracts or chains.
05	**Interrupted transitions**	Order mint, burn and unlock so a failed step cannot strand or duplicate value.	A burn without an unlock, or an unlock while supply still exists.

`ETH_GETPROOF / -32601` The testnet EVM endpoint does not implement account proofs. The EVM witness path needs a real source before rule one can be enforced. We report the node's refusal rather than invent a witness.

Enforcement lives in the receiving runtime. A message is never the authority. Root source today is a trusted finalized checkpoint, and removing that trust is part of the work.

WHAT IS LIVE TODAY

A console, a client-side witness, and no deployed title.

Every figure below came from the two public endpoints on 2026-11-02 at 14:20 UTC. Nothing here is a projection.
The contract layer has 71 Foundry tests; the proof service has 1,077 tests.

BEST HEAD	FINALIZED HEAD	FINALITY LAG	REVIVE HEAD	CHAIN ID
292,823	292,820	3	292,823	420420777
chain_getHeader	state root 0x77db…4e53	blocks behind best	eth_blockNumber	eth_chainId, genesis matched

STATUS / COMPONENT	EVIDENCE
Native backing and ownership registry	IMPLEMENTATION REQUIRED
ERC-20 / ERC-721 representation	TESTED, NOT DEPLOYED
Native to EVM state proof verification	TESTED
EVM to native state proof verification	BLOCKED BY MODULE
Revision and replay protection	TESTED, NOT DEPLOYED
End-to-end deployment	NOT DEMONSTRATED
Security audit	NOT PERFORMED

CONSOLE — NATIVE + REVIVE

Finalized heads, finality lag and slot cadence from the native node. Head and chain id from Revive. Plain JSON-RPC from the browser, serialized to survive the public endpoints.

/console /console/network

NATIVE WITNESS — VERIFIED IN THE BROWSER

A storage proof for System.Number at the finalized head. Every trie node is re-hashed with blake2b-256 and checked against the state root the chain published.

/console/proofs

PROOF SERVICE — NOT DEPLOYED

Trie verifier, typed RPC clients and a Fastify API, tested with 1,077 cases including a live captured proof. The contract layer adds 71 Foundry tests. No endpoint is deployed.

server/ / contracts/ / fixtures

ETH_GETPROOF REFUSED — Revive answered -32601. Root source today is a trusted finalized checkpoint.

WHY PORTALDOT

This model needs both runtimes. Portaldot has one live and one listed.

NETWORK	WHAT IT GIVES THE ASSET	WHAT IS MISSING
Portaldot today / Revive only	ERC-20 and ERC-721 markets, Solidity applications, live finality, and native assets, identity, vesting and proxy records in the runtime.	No WASM contracts module, so no on-chain native title for a verifier to read.
WASM contracts only, elsewhere	Contract-level native ownership, identity, vesting and permission rules.	No EVM market. The representation has nowhere to execute.
Portaldot with both	Revive live now, WASM contracts module for V3.0, one finality. Proofs can cross once both exist.	The native gate cannot be enforced until that module ships.

The asset does not choose between WASM and EVM. It belongs to both, and neither can act without proving that the other agrees.

LEMO DESIGN INVARIANT / NOT YET ENFORCED END TO END

> **No cross-call dependency.** ink! to Solidity calls are awaiting confirmation. The design moves proof bytes instead, and carrying them authorizes nothing.

The EVM-side gate is buildable now. The native-side check waits on the module. Statement Store stays a noticeboard: signed announcements, never authorization.

SHARED FINALITY

The proof and the state it proves settle under one validator set.

TWO VERIFIERS

One on EVM now, one native when the module ships.

NO RELAYER AUTHORITY

A relayer can carry bytes. It cannot make a bad proof verify.

THE ASK

Build the EVM gate now. Build the native gate when the module ships.

The design is public and the console is live. What is missing is the deployed pair: an EVM gate with a real title source, a native gate once the WASM contracts module lands, and the rejection cases proven on chain.

M1 / DEPENDENCY

Module and noticeboard

Confirm the WASM contracts module on V3 testnet, and Statement Store support on the target node.

M2 / TITLE AND GATE

Registry plus EVM verifier

Title registry with real backing. EVM verifier deployed against the same title id and revision.

M3 / BOTH DIRECTIONS

Two verifiers, two roots

Native state proven on EVM, EVM state proven on native, each against a finalized head.

M4 / REJECTIONS

Break it on purpose

Cleared identity, unbound backing, stale or replayed proof, outstanding supply. Then redeem and unlock.

WHAT WE NEED

Engineering time to land M2 to M4, a security review before any real asset uses this, and a Portaldot contact to confirm the WASM module and Statement Store timelines. No capital ask, no token sale, no fake traction.

No registry. No token. No audit. Everything here is design, live network state, or a named dependency, and every slide says which.

REPOSITORY

github.com/subheeksh5599/lemo